S.Y. B. Tech. Academic Year 2026-27 Semester: III

CSE20350/AID20210/CLD 20020: Data Structures Laboratory

WRITE UP

Experiment Number: 09

TITLE: Printer Queue Manager

PROBLEM STATEMENT:

Write a C program to simulate a Printer Queue Manager using a circular queue implemented
with an array. a) Add a Print Job (Enqueue) b) Process a Print Job (Dequeue) c) Display All
Pending Print Jobs

OBJECTIVE:

1. To implement a Circular Queue for managing print jobs efficiently using the FIFO (First-In, First-Out) principle.
2. To perform queue operations such as Enqueue, Dequeue, and Display while handling overflow and underflow conditions.
3. To demonstrate efficient memory utilization by reusing vacant queue positions through circular queue implementation in a networked printer management system.

THEORY:

A Circular Queue is a special type of queue in which the last position is connected back to the first position, forming a circle. It follows the FIFO (First In, First Out) principle, but unlike a simple queue, it efficiently utilizes the available memory by reusing the empty spaces created after deletion.

Need for circular queue:

In a simple (linear) queue, once the REAR reaches the last index of the array, no more insertions are possible, even if there are empty positions at the beginning

due to deletions. Circular Queue solves this problem by wrapping the REAR to the beginning of the array.

Features of Circular Queue

- Follows the FIFO (First In, First Out) principle.
- Last position connects to the first position.
- Efficient utilization of memory.
- Insertion at Rear and deletion at Front.

IMPLEMENTATION:

- **PLATFORM:**
  - 64-bit Open source Linux or its derivatives.
  - Open Source C Programming tool like gcc/Eclipse Editor.

- **TEST CONDITIONS:-**

Test Case	Input	Expected Output
TC01	Create an empty circular queue	Queue is initialized successfully.
TC02	Enqueue Job ID: 101, Title: "Report.pdf"	Print job is added successfully to the queue.
TC03	Enqueue multiple print jobs until queue is full	All jobs are inserted successfully until the queue reaches maximum capacity.
TC04	Enqueue one more job when queue is full	Display "Queue Overflow! Print queue is full." No job is inserted.
TC05	Display queue after inserting jobs	All pending print jobs are displayed in FIFO order with Job ID and Document Title.
TC06	Dequeue one print job	The first inserted print job is removed and displayed as processed.
TC07	Dequeue all print jobs	All jobs are processed one by one in FIFO order until the queue becomes empty.
TC08	Dequeue from an empty queue	Display "Queue Underflow! No print jobs available."
TC09	Enqueue a new job after dequeuing some jobs (wrap-	New print job is inserted into the vacant position created by earlier deletions,

	around condition)	demonstrating circular memory reuse.
TC10	Display queue after wrap-around insertion	Queue displays all remaining and newly added jobs in the correct FIFO sequence.
TC11	Alternate Enqueue and Dequeue operations	Queue maintains the correct order of print jobs without data loss or incorrect indexing.
TC12	Exit the program	Program terminates successfully.

- **PSEUDO CODE:** ***//To be Written by Students***

  Write pseudocode for initialization, insertion (enqueue), deletion (dequeue), display, overflow checking, and underflow checking using a Circular Queue.

- **TIME COMPLEXITY:** ***//To be Written by Students***

  Write time complexity of above operations

- **CONCLUSION:**

  The Circular Queue efficiently manages print jobs by processing them in **First-In, First-Out (FIFO)** order while reusing memory through circular indexing. It minimizes memory wastage, improves resource utilization, and provides an effective solution for real-world printer management systems.

FAQs ***//To be Written by Students***

1. What is a Circular Queue?
2. Why is a Circular Queue preferred over a Linear Queue?
3. What principle does a Circular Queue follow?
4. What are the basic operations performed on a Circular Queue?
5. What is an overflow condition in a Circular Queue?
6. What is an underflow condition in a Circular Queue?
7. How does a Circular Queue reuse memory?
8. What are the applications of a Circular Queue?
9. What is the time complexity of Enqueue and Dequeue operations?
10. How is a Circular Queue implemented in C?

Practice Assignments – Circular Queue

1. **Student Registration System**
   Develop a Circular Queue program to manage student registration requests. Implement **Enqueue**, **Dequeue**, and **Display** operations, and handle overflow and underflow conditions.

2. **Vehicle Parking Management**
   Design a Circular Queue to manage the entry and exit of vehicles in a parking area. Display the current queue of vehicles and demonstrate memory reuse after vehicles leave.

3. **Hospital Token Management System**
   Implement a Circular Queue to manage patient tokens. Process patients in FIFO order and display the waiting list after each operation.

4. **Call Center Customer Queue**
   Create a menu-driven Circular Queue application to manage incoming customer calls. Implement insertion, deletion, display, and queue status operations while efficiently utilizing memory.

5. **Food Order Processing System**
   Develop a Circular Queue to manage food orders in a restaurant. Each order should contain an **Order ID** and **Customer Name**. Process orders in FIFO order, display pending orders, and handle queue overflow and underflow situations.